Final Report – Fan Blade Redesigning

Problem Statement and background

The Maverick Coastal 60-inch Outdoor Ceiling Fan which can be seen in figure 1, produced by Visual Comfort & Co., is designed for outdoor residential use. This model features a three-blade configuration and is available in white and black finishes. Approximately 6,400 units were sold in the United States, with an additional 200 units in Canada, through various retailers like Home Depot, from January 2021 through November 2023, priced at approximately $850. The component of concern is the fan blade. There have been nine reported incidents of the fan blades cracking or breaking during operation, posing a significant impact injury hazard if the blades detach and fall. Although no injuries or property damage have been reported to date, the potential risk necessitates a redesign of the fan blades to enhance their structural integrity and ensure user safety [1].

Ceiling fan blades are critical components responsible for generating airflow in residential and commercial spaces. Their primary function is to convert the rotational motion of the fan motor into air circulation, thereby enhancing thermal comfort by promoting evaporative cooling and improving air distribution. The aerodynamic shape of the blades—often resembling an airfoil—creates a pressure difference between the upper and lower surfaces during rotation, which produces airflow. This airflow direction and intensity depend heavily on the blade angle (pitch), length, material stiffness, and rotation speed. For larger fans, especially those with long or slender blades, the centrifugal force during rotation becomes a significant mechanical load. As such, blades must be engineered not only for aerodynamic performance but also for structural integrity under dynamic loading [2].

Figure 1. Image of the Recalled Fan [1]

The materials used for ceiling fan blades vary, including wood, metal, and plastic, each offering different benefits. Wooden blades provide a natural aesthetic but may require protective coatings for durability. Metal blades are durable and suitable for modern designs but can be heavier, affecting energy efficiency. Plastic blades are lightweight and versatile, making them suitable for both indoor and outdoor use. For outdoor ceiling fans, materials must withstand environmental factors such as humidity and UV exposure; thus, using weather-resistant materials and UV-protective coatings is essential for longevity [2].

The primary impetus for the redesign is to eliminate the risk of blade failure, thereby ensuring the safety and reliability of the product. Key factors to consider in the redesign include selecting materials with superior strength and durability to withstand operational stresses and environmental conditions, implementing manufacturing techniques that enhance the structural integrity of the blades, and maintaining the visual design elements that attract customers, ensuring that the redesigned blades align with the overall aesthetic of the fan. Cost efficiency is also crucial, balancing improved safety features with cost-effective production to keep the product competitively priced. By addressing these factors, the redesigned fan aims to restore consumer confidence, comply with safety standards, and maintain market competitiveness.

Translation

Material and process selection is being carried out on a ceiling fan blade to address the performance shortfall of potential crack in blade and impact injury hazards. To maximize safety and ensure reliable operation, it is necessary to optimize the blade's structural integrity and durability. Structural integrity will be maximized from the standpoint of fatigue, as how and fracture toughness, as fan blades must withstand continuous rotational forces, vibration, and occasional impacts without failing. Below are blade details which will be used during the blade redesigning.

Blade Details:

Blade Group	Diameter	Included	RPM Hi	CFM Hi	Watts Hi	Amps Hi	RPM Med	CFM Med	Watts Med	Amps Med	RPM Low	CFM Low	Watts Low	Amps Low
	60	Yes	180.0	8574.0	26.7	0.3					59.0	3500.0	2.8	0.04

Figure 2. Specifications of Fan Blade [3]

Ceiling fan blades are subjected to wear and tear in service, with failure most likely to occur due to fatigue from cyclic loading, environmental degradation, or improper installation. For this reason, a key performance objective is to maximize the blade's resistance to fatigue, corrosion and durability while maintaining a lightweight design to reduce stress on the motor. To best capture the service conditions and potential failure modes of a ceiling fan blade, material selection will prioritize high fracture toughness, and stiffness. The main constraints and objectives for the fan blade material and process selection are shown in (Table 1).

Table 1. Translation for Fan Blade

Function: To provide efficient air circulation while ensuring structural integrity under operational loads. (rotating blade)

Constraints	Young's Modulus (Minimum)	0.14 GPa
	Yield Strength (Minimum)	0.376 MPa
	Fracture Toughness (Minimum)	1.52 MPa-$m^{1/2}$
	Maximum Service Temperature (minimum)	40 °C
	Minimum Service Temperature (maximum)	-40 °C
	Material Price (Maximum)	$5/kg
	Batch size (Minimum)	20000 Units
	Durability - UV Radiation	Good or Excellent
	Shape	3D Solid

Objective	Material Index
Maximize resistance to fast fracture; blade length fixed	K_{1c} / ρ
Maximize resistance to centrifugal loading; blade length fixed	σ_e / ρ

Free Variable: Blade thickness, choice of material, choice of process

Based on the established constraints, a systematic approach involving material index screening and ranking will be employed to identify suitable materials for the redesigned fan blade, followed by detailed process optimization and further analysis.

Screening

The goal of the screening process is to identify materials that satisfy the minimum performance requirements necessary for the redesigned outdoor ceiling fan blade. The failed product utilized Acrylonitrile Styrene Acrylate (ASA), which exhibited insufficient fracture resistance under operational stresses. To prevent similar failures, this redesign targets improved mechanical performance particularly resistance to fracture and fatigue while maintaining manufacturability, environmental durability, and cost-effectiveness.

To shortlist of potential material candidates for the redesigned ceiling fan blade, the constraints finalized in translation stage were applied to the Level 3 database within Granta Edu Pack software[4]. These constraints were used individually at first to assess the effect of each one independently and to ensure that no single constraint overly restricted the range of potential options. The screening focused specifically on bulk materials compatible that would be practical for manufacturing.

Initially, mechanical and thermal constraints were prioritized to ensure that any selected material would perform safely under operational loads. The constraints included a minimum Young's Modulus of 0.14 GPa, Yield Strength of 0.376 MPa, and Fracture Toughness of at least 1.52 MPa-m^1/2, equivalent to the average performance of ASA, the failed material. Additionally, candidate materials were required to withstand service temperatures ranging from -20°C to 40°C. A cost ceiling of $5/kg was also applied to preserve manufacturing feasibility at the target production volume of 20,000 units. At this stage, durability constraints such as UV resistance were not enforced, to avoid prematurely eliminating materials that could potentially be improved through surface treatments, such as UV-resistant coatings or additive stabilizers[5]. This approach preserved flexibility in later stages of selection, where trade-offs between base material performance and post-processing options could be more thoroughly evaluated.

After applying all non-durability constraints, approximately 1,285 materials remained in the candidate pool. This filtered set included a diverse range of thermoplastic polymers such as ABS and polycarbonates, along with various lightweight metals including aluminum and magnesium alloys, as well as a broad selection of steels. The wide variety of materials retained offered meaningful diversity in terms of density, mechanical strength, fracture toughness, and cost, which is essential for a thorough and well-balanced evaluation. Given the breadth and quality of the remaining options, the initial constraint set was deemed appropriately selective and did not require further adjustment. The materials carried forward at this stage represented viable candidates for outdoor fan blade applications and were suitable for more detailed analysis in the subsequent ranking process.

To evaluate the impact of environmental durability, the UV resistance constraint was temporarily applied, which resulted in the removal of 215 materials, the majority of which were various grades of ABS. While applying this filter still left a sufficient pool of candidates for further analysis, the decision was made to exclude durability as a strict constraint during the initial screening. This approach was taken to avoid prematurely eliminating otherwise promising materials that could be enhanced through surface treatments, such as UV-stabilizing coatings or additive integration. The potential need for such treatments will be carefully reconsidered during the ranking phase, particularly if one of the top-performing materials lacks inherent UV resistance but excels in other critical performance areas[5].

Following this initial material screening, indices representing the design objectives were used to narrow the list further. On each chart selection line was used to narrow down to 9 common material candidates using the bubble plots for two materials indices. For narrowed list of choices, long list of aluminum, magnesium and various steel alloys were consolidated and represented by selective material within that family with highest material index. For example, aluminum alloy was Aluminum 7475, T7351 was chosen.

Following this initial material screening, a refined selection was conducted. Selection lines were applied to visually filter materials, reducing the candidate pool down to 9 materials that best aligned with the project's design needs. As part of this refinement, groups of similar materials such as those within the aluminum, magnesium, and steel families—were consolidated to avoid redundancy in the ranking phase. Instead of carrying forward every variation of a given alloy type, a single representative material from each family was selected based on its superior overall performance. For example, within the aluminum alloys, Aluminum 7457-T7351 was chosen as the most promising option. This approach ensured a balanced and diverse shortlist of candidates, while maintaining clarity and focus on ranking.

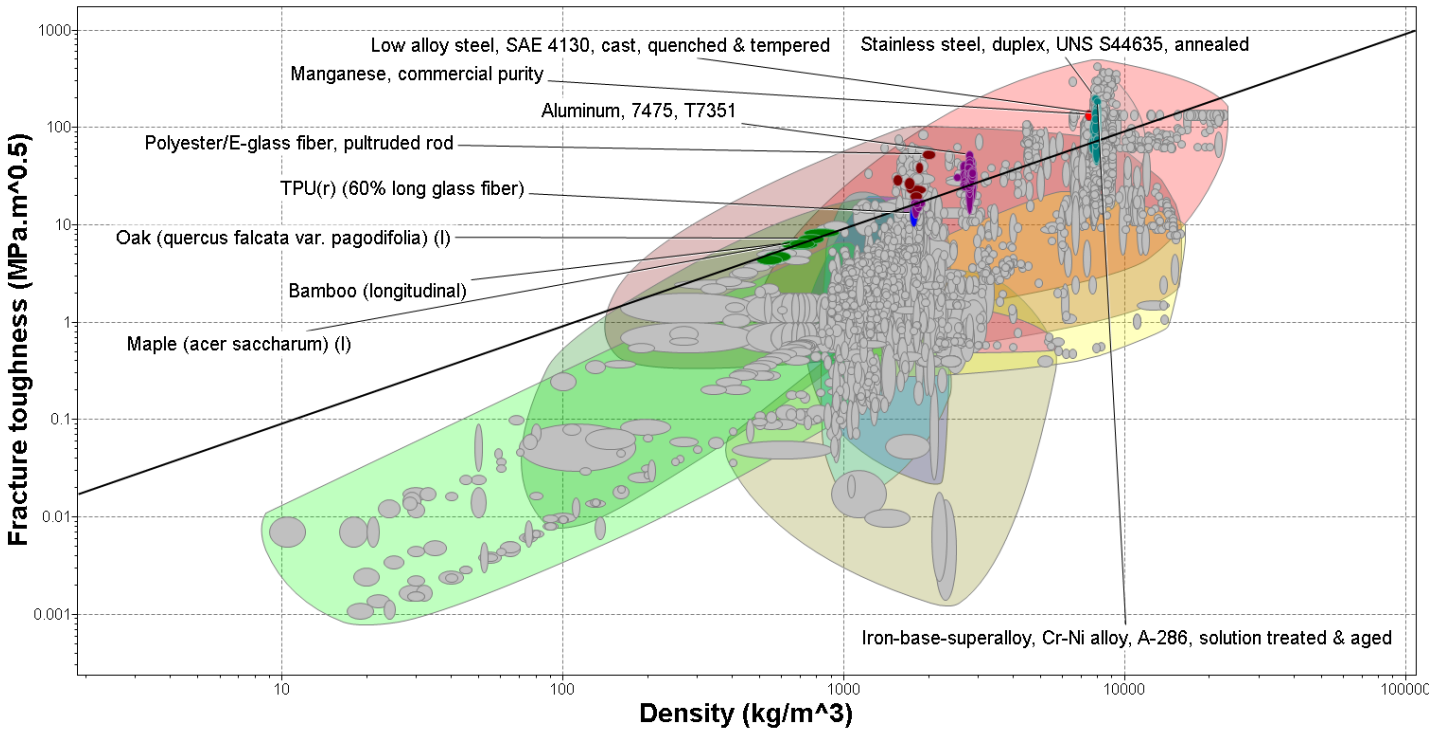


Figure 3. Chart maximizing the material index K1c/ρ for resistance to fracture in damage tolerant design. Top materials have been labeled

Fracture toughness (K1c) is the most important property to consider in this application, as the primary failure mode of the original fan blade was cracking and eventual breakage under operational stress. Figure 3 shows a plot of fracture toughness vs. density, which was used to identify materials that offer both high resistances to crack propagation and low weight, an ideal combination for rotating components like ceiling fan blades. In this plot, materials such as aluminum alloys (e.g., 7475,T7351) and composites like glass-fiber reinforced polyester occupy favorable regions and are one of the top choices according to this index, balancing moderate density with significantly higher toughness than unreinforced plastics. These materials are more likely to prevent sudden failure and improve long-term durability. Although some high-performance metals such as steel alloys show excellent toughness, their density and cost make them impractical for this consumer product.

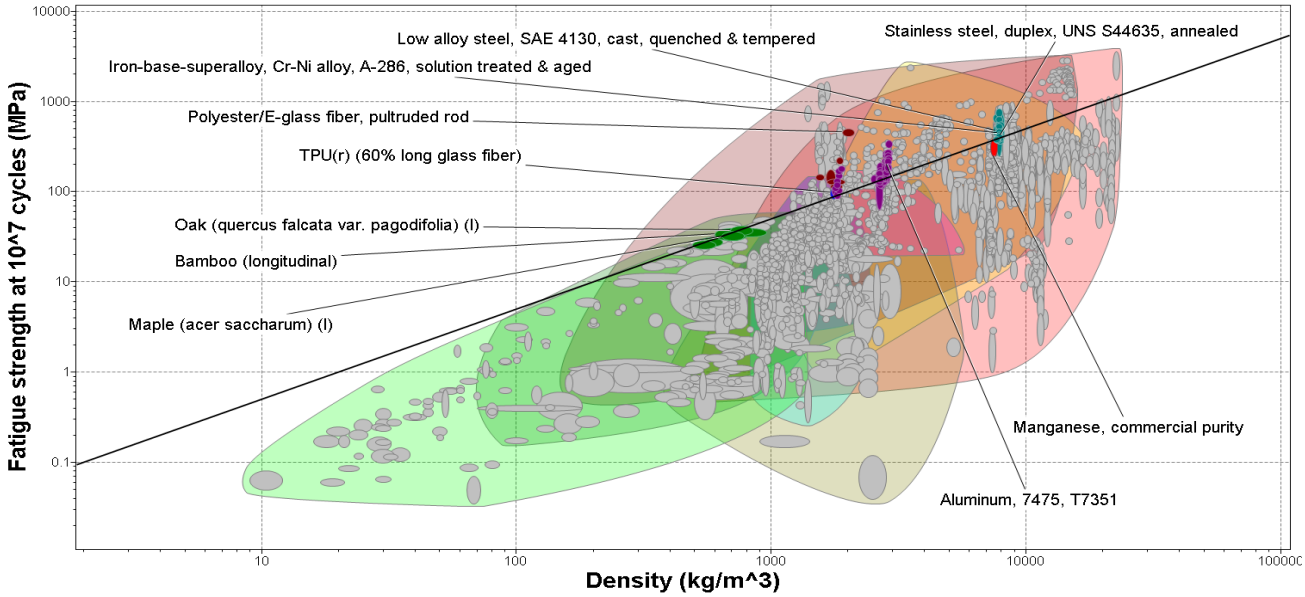


Figure 4. Chart maximizing the material index σ_e/ρ for resistance to centrifugal loading in damage tolerant design.

The second most important consideration in material selection for the fan blade is its ability to withstand centrifugal loading during rotation, which directly relates to fatigue strength under cyclic stresses (endurance limit), as the fan blades has cyclic loading. Figure 4 presents a plot of fatigue strength at 10^7 cycles vs. density, highlighting materials that can maintain structural integrity under repeated loading without initiating cracks or deformation. This is particularly important for ceiling fan blades, which experience continuous rotational motion over extended periods. Materials located above the diagonal line on the plot offer a favorable combination of high fatigue strength and low density, both of which reduce stress on the motor and extend the product's service life. Notably, materials such as aluminum alloys and magnesium alloys exhibit strong performance in this region. While some high-strength steels also meet these criteria, their higher density may increase system loading and noise.

From the three charts above, selected materials were shortlisted which are noted in table 2.

Table 2. Shortlisted materials which will be ranked based on the weighted performance (listed in alphabetical order)

Material	Family
Aluminum, 7475, T7351, Wrought	Metals, non-ferrous
Bamboo longitudinal	Natural, wood
Iron-base-superalloy, Cr-Ni alloy, A-286, soln. treated & aged	Metal, ferrous
Low alloy steel, 4130, cast, hardened and tempered	Metal, non-ferrous
Manganese, commercial purity	Metal, others
Maple (acer saccharum), longitudinal direction	Natural, wood
Oak, longitudinal direction	Natural, woods
Polyester/E-glass Fiber pultruded rod	Composites, polymer matrix
Stainless steel, duplex, UNS S44635, annealed	Metal, ferrous
TPU(r), (60% long glass fiber)	Plastic, thermoplastic, amorphous

Ranking

To determine the most suitable material from the shortlisted candidates for the blade, a quantitative ranking procedure was employed. This approach involved assigning weight factors to various performance criteria to objectively evaluate each material's suitability. From the initial screening step in this process top candidate materials were obtained. Material properties of these materials were then used to calculate specific material indices to the selection study. By quantifying the performance of each material against the defined criteria, this method facilitated a systematic comparison, ultimately guiding the selection of the optimal material for the intended application.

Table 3. Material Properties (lower end) of Shortlisted Materials [5]

Material	Material Family	ρ (kg/m^3)	Cm (USD/kg)	K1c (MPa*m^0.5)	σe (Gpa)
Aluminum, 7475, T7351, Wrought	Metals, non-ferrous	2.78E+03	3.89	46.4	205
Bamboo longitudinal	Natural, wood	600	1.34	5.7	29.3
Iron-base-superalloy, Cr-Ni alloy, A-286, soln. treated & aged	Metal, ferrous	7.90E+03	4.51	180	407
Low alloy steel, 4130, cast, hardened and tempered	Metal, non-ferrous	7.81E+03	0.894	122	445
Manganese, commercial purity	Metal, others	7.35E+03	1.8	120	250
Maple (acer saccharum), longitudinal direction	Natural, wood	640	1.34	5.8	29.4
Oak, longitudinal direction	Natural, woods	690	0.66	6.5	33.7
Polyester/E-glass Fiber pultruded rod	Composites, polymer matrix	1.90E+03	1.75	48.4	414
Stainless steel, duplex, UNS S44635, annealed	Metal, ferrous	7.72E+03	3.43	192	361

Using the material properties from table 3, the material indices were calculated (M1 & M2) in table 5. In this calculation the units Since the goal of this step was to enable a relative comparison between materials, the specific units were not considered, as they remained consistent across all entries. Once calculated, each index value was normalized by the maximum value of that index among the shortlisted materials. For example, all K_{1c}/ρ values were divided by the highest K_{1c}/ρ value in the list to produce a normalized toughness index. This approach provided a dimensionless comparison matrix (N1 & N2) that enabled objective evaluation across different criteria. The resulting normalized values were then compiled, shown in table 4, forming the basis for the weighted factor analysis which was used to further rank the materials.

Table 4. Material index and normalized material indices for shortlisted materials

M1 (K1c/ρ)	M2 (σe/ρ)	N1	N2
0.02	0.07	0.6552114	0.3384249
0.01	0.05	0.3729339	0.2241143
0.02	0.05	0.894445	0.2364398
0.02	0.06	0.613221	0.2614943
0.02	0.03	0.6409175	0.1561011
0.01	0.05	0.3557593	0.2108243
0.01	0.05	0.3698048	0.2241476
0.03	0.22	1	1
0.02	0.05	0.97632	0.2146066

To combine each of the normalized material indices into a single score for ranking the candidate materials, weight factors were assigned based on the most critical requirements of the fan blade redesign. In this study, two material indices were considered equally important. The first was the fracture toughness-to-density ratio (K1c /ρ), which reflects a material's ability to resist crack propagation while keeping the blade lightweight. This is crucial, as the original failure mode involved blade cracking and detachment. The second index was the fatigue strength-to-density ratio (σe/ρ), which is also essential due to the high rotational speeds the blades experience during operation. Materials must endure continuous cyclic loading without undergoing plastic deformation, and a higher σe/ρ helps reduce the load on the motor by minimizing weight. For these reasons, K1c/ρ and σe/ρ were each assigned a weight factor of 0.6 and 0.4 respectively, ensuring a balanced evaluation of both fracture resistance and structural strength in the final material selection.

The weight factor has been summarized in (Table 5).

Table 5. Weight factor for ranking analysis

Material Index	Weight Factor
M1 (K1c/ρ)	0.6
M2 (σe/ρ)	0.4

Using these weight factors to multiply the corresponding normalized material index for each material entry and then adding together the result yields the weighted material index. The value of this derived index directly correlates to the suitability of the material to the design goals defined. These derived indices are summarized below (Table 6). These entries are rearranged so that they are ranked from max weighted index to min which clearly shows the order of most suitable material choice to least.

Table 6. Ranking of shortlisted materials

Material	Material Family	Mw
Polyester/E-glass Fiber pultruded rod	Composites, polymer matrix	1
Stainless steel, duplex, UNS S44635, annealed	Metal, ferrous	0.671634633
Iron-base-superalloy, Cr-Ni alloy, A-286, soln. treated & aged	Metal, ferrous	0.631242934
Aluminum, 7475, T7351, Wrought	Metals, non-ferrous	0.528496786
Low alloy steel, 4130, cast, hardened and tempered	Metal, non-ferrous	0.472530337
Manganese, commercial purity	Metal, others	0.44699095
Bamboo longitudinal	Natural, wood	0.297785289
Oak, longitudinal direction	Natural, woods	0.311541895
Maple (acer saccharum), longitudinal direction	Natural, wood	0.297785289

Although this material ranking gives good indication of suitability there are other aspects of that make some materials more desirable than others like process compatibility, cost considerations, etc. These will be discussed in detail way in next section.

Documentation

While material indices are invaluable for narrowing down potential candidates, they serve as an initial filter. After this ranking, it's essential to delve deeper into the top-performing materials, examining their specific properties, manufacturability, and cost-effectiveness. This comprehensive evaluation, informed by data from Granta Edu Pack[5], ensures that the final material selection aligns with practical considerations beyond the preliminary indices. Such an approach facilitates a well-rounded decision-making process, balancing performance metrics with real-world applicability.

Using the top candidate materials that were shortlisted based on material index, a holistic view of these materials needs to be provided, to finalize the choice of material for fan blade. The top candidates are Polyester/E-glass Fiber pultruded rod, Stainless steel duplex (UNS S44635, annealed), Iron-base-superalloy Cr-Ni alloy (A-286, soln. treated & aged), Aluminum 7475 T7351 Wrought. Figure 5 and Figure 6 provides bar plot for fracture toughness (K1c) and cost per unit volume (USD/m^3) with the selected materials labeled.

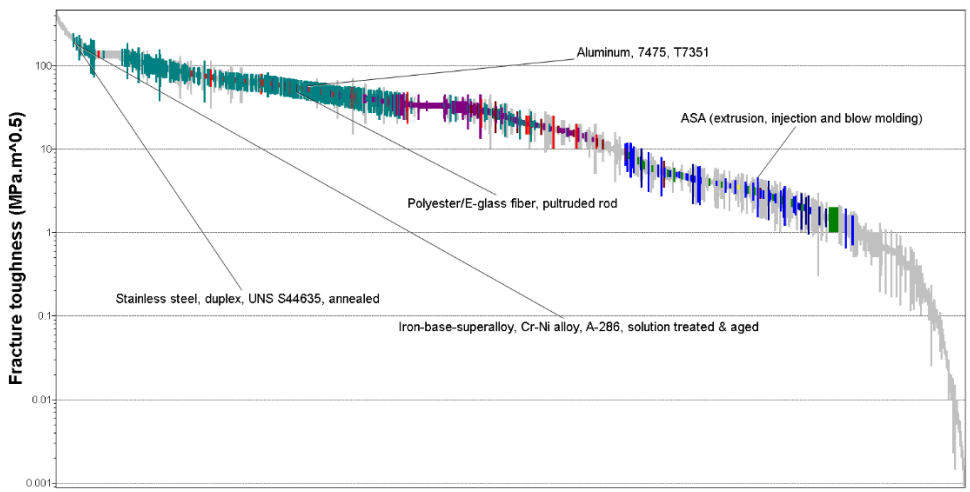


Figure 5. Bar plot for Fracture Toughness (K1c) of various materials

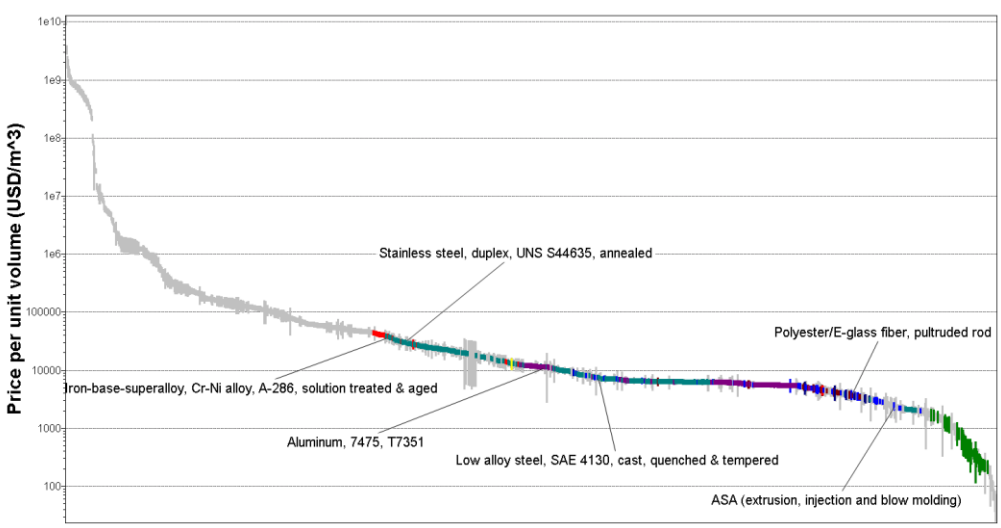


Figure 6. Bar plot for Price per unit volume of material

These four materials are evaluated based on fracture toughness, price per unit volume and manufacturability.

1. **Polyester/E-glass Fiber, pultruded rod:** The polyester/E-glass fiber pultruded rod provides an optimal balance of moderate density (1900 kg/m³) and high mechanical performance, making it an excellent candidate for rotating fan blades. Its fracture toughness (48.4 MPa·m^0.5) exceeds that of aluminum, ensuring superior resistance to crack propagation—critical for addressing the original ASA blades' failure modes. The fatigue strength (414 MPa) is exceptionally high, ideal for continuous rotation.
   For manufacturing, pultrusion is suitable for straight profiles, while compression molding of Sheet Molding Compound (SMC) is preferred for 3D airfoil shapes. For large production runs with desirable shapes, compression molding is especially favorable, allowing consistent part quality, short cycle times, and less labor intensive. The process is already widely used in the electrical components and housing appliances. A UV-resistant coating is required for outdoor use, as polyester resins degrade under prolonged sunlight[Table 4].

2. **Stainless steel, duplex, UNS S44635, annealed:** Duplex stainless steel (UNS S44635) offers exceptionally high mechanical durability, with a fracture toughness of 192 MPa-m^0.5, the highest among all materials listed, which can be seen in figure 5. This indicates superior resistance to crack initiation and propagation, especially in applications involving sharp stress concentrations or impact. The material's fatigue strength of 361 MPa is also robust, suggesting long service life under repeated loading. However, its density of 7720 kg/m³ is among the highest, which poses challenges for use in rotating blades. The additional weight increases the centrifugal force experienced during operation and can lead to increased motor wear and potential safety hazards if not carefully balanced.
   While stainless steel offers excellent strength and durability, its high density approximately 7,930 to 8,000 kg/m³ for grades like 304 and 316 makes it significantly heavier than alternative materials such as fiber-reinforced composites. In the context of ceiling fan blades, this increased weight would impose a greater load on the motor, necessitating more power to achieve and maintain desired rotational speeds. Consequently, using stainless steel for fan blades could lead to higher energy consumption and reduced motor efficiency. Therefore, despite its favorable mechanical properties, stainless steel cannot be considered as lighter materials can provide sufficient strength while enhancing overall performance and energy efficiency[5].

3. **Iron-base-superalloy, Cr-Ni alloy, A-286, soln. treated & aged:** A-286 superalloy offers exceptional mechanical performance, with a fracture toughness of 180 MPa-m^0.5 and fatigue strength of 407 MPa, both near the top of the range among all listed candidates. These values make it extremely resistant to both crack growth and fatigue failure, a reason why it's commonly used in aerospace components like jet engine fasteners and turbine discs. However, it shares a major drawback with stainless steel: a very high density of 7900 kg/m³, which makes it one of the heaviest materials considered. This high mass would increase the loading on the motor and mounting system significantly, making it difficult to maintain the energy efficiency and balance expected from consumer-grade ceiling fans.
   Moreover, A-286's manufacturing challenges stemming from its toughness and heat resistance make it difficult to machine, forge, or stamp, leading to higher production costs and longer fabrication times . These factors render A-286 over-engineered for consumer-grade ceiling fan blades, where cost-effectiveness and ease of manufacturing are critical. Therefore, despite its superior mechanical properties, A-286 is not a practical choice for this application[Table 5].

4. **Aluminum, 7475, T7351, Wrought:** Aluminum 7475-T7351 stands out for its low-density of 2780 kg/m³ one of the lightest structural metals offering significant weight savings compared to ferrous options. Its fracture toughness is very close to that of the fiberglass composite as shown in figure 5, reflecting good damage tolerance and the ability to resist crack propagation under stress but rather less compared to its alternatives. In terms of fatigue performance, it has a strength of 205 MPa, which is considerably lower than the composites and steels but still adequate for moderate cyclic stress scenarios like fan blade operation. This lower fatigue value could potentially be offset by increased blade thickness or geometry optimization.
   For high-volume production, aluminum is highly adaptable. Stamping or forging (hot or cold closed die forging) are suitable manufacturing methods, especially when dealing with wrought grades like 7475. Forging provides enhanced grain structure and mechanical strength, while stamping allows rapid production at relatively low cost per part. These processes are compatible with large-scale output, although surface protection such as anodizing or powder coating is necessary to prevent corrosion in outdoor environments. Overall, aluminum 7475 is a practical and lightweight option with solid structural performance and proven manufacturability, though it slightly underperforms in fatigue resistance compared to composites or steel. So, It can be considered for final material selections[Table 5].

Final Selection

Material Selection

After thorough evaluation of mechanical properties, cost, and manufacturing feasibility, the final material selected for the redesigned ceiling fan blade is Aluminum Alloy 7475-T7351. While Polyester/E-glass fiber composite in pultruded rod form offers superior fracture toughness up to 59.2 MPa·m^0.5 and fatigue strength ~497 MPa [Table 4], it exhibits strong anisotropy due to its unidirectional fiber alignment, making it highly resistant in one direction but vulnerable in shear and off-axis loading[7]. This directional dependency poses a risk in applications like fan blades, where multi-axial stress and cyclic loading are prevalent leading to rejection of the composite as ideal choice manufacturing. Which leaves us with Aluminum alloy is the final material for selection[8].

This significantly outperforms ASA (Acrylonitrile Styrene Acrylate) the polymer originally used in the recalled product in all key mechanical performance criteria relevant to ceiling fan blade applications. In terms of fracture toughness, Aluminum 7475-T7351 offers a value of 46.4 MPa·m^0.5, compared to just 1.52 MPa·m^0.5 for ASA. As shown in Figure 5, ASA appears near the bottom of the toughness range, while aluminum is positioned much higher, demonstrating vastly superior resistance to crack initiation and propagation under dynamic and impact loads. From a fatigue and strength standpoint, Aluminum 7475-T7351 again demonstrates clear superiority. It provides a fatigue strength of 205 MPa and tensile strength well above 450 MPa, whereas ASA's fatigue strength is considerably lower, and tensile strength ranges only between 55 to 65 MPa.

Although aluminum has a higher density (2780 kg/m³) than ASA (~1060 kg/m³), its superior strength-to-weight ratio ensures that the fan blade remains structurally efficient without significantly overloading the motor. Additionally, its isotropic mechanical properties make it more reliable in handling multi-directional stresses compared to polymers or composites with directional strength. Economically, as seen in Figure 6, aluminum is moderately more expensive per unit volume than ASA, but it remains cost-effective for high-volume manufacturing due to its compatibility with efficient shaping processes like forging and machining. ASA may offer lower raw material costs, but its poor durability led to real-world failures and product recalls. In contrast, Aluminum 7475-T7351 ensures a substantial performance upgrade, providing long-term safety, reliability, and manufacturability without compromising economic viability[11][12].

Aluminum 7475-T7351 is a high-strength, heat-treatable wrought alloy belonging to the 7000 series, primarily alloyed with zinc, magnesium, and copper. The "7475" designation indicates a variant of the 7075 alloy, modified to enhance fracture toughness and stress corrosion resistance through controlled impurity levels and the addition of elements like chromium . As a wrought alloy, 7475 is processed through methods such as rolling, extrusion, or forging, resulting in improved grain structure and mechanical reliability compared to cast alloys. [11]

The "T7351" temper designation follows the Aluminum Association's standard nomenclature, indicating a specific sequence of thermal and mechanical treatments:

- **T7**: Solution heat-treated and overaged to achieve a stable condition, enhancing stress corrosion resistance.
- **3**: Stress-relieved by stretching, typically around 1.5–3%, to reduce residual stresses from fabrication processes.
- **51**: Indicates that the product was stress-relieved by stretching and then artificially aged. [12]

This tempering process results in a material with balanced properties suitable for applications requiring high strength, fracture toughness, and resistance to stress corrosion cracking, such as aerospace components and, in this context, high-performance outdoor fan blades.

Primary Shaping Process

Since, final material for manufacturing of fan blade has been decided an apt primary shaping process is required to be selected for it. To reduce the process, few constraints were applied (such as 3-D solid shape, economic batch size limit of minimum 20000 units [Table 1] low & medium labor intensity and primary shaping as process characteristic). After the limits were applied a bar plot of Shaping process and Economic batch size was prepared and few manufacturing process suitable for Aluminum alloy were down selected which can be observed in figure 7.

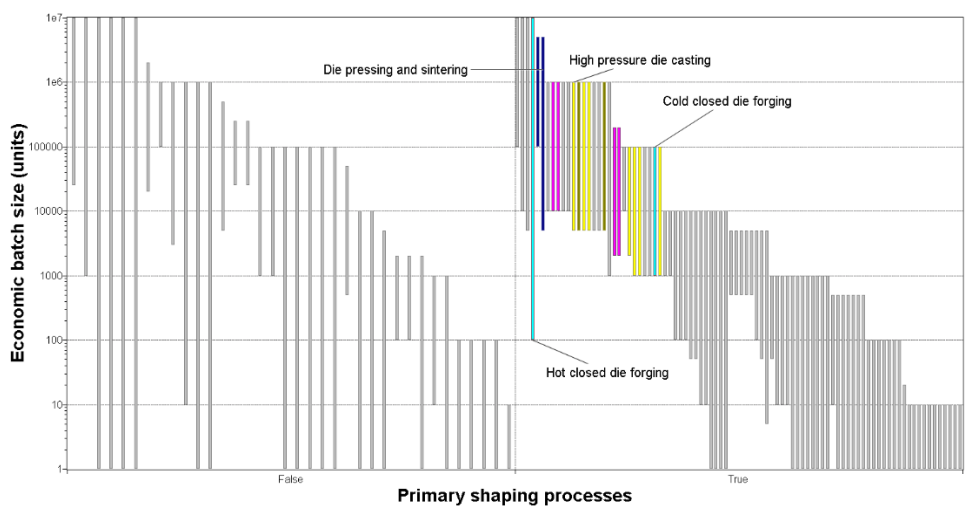


Figure 7. Bar plot for primary shaping process plotted against economic batch size, with set limit constraints

Cold closed-die forging was identified as the most appropriate primary shaping process for the 7475-T7351aluminum alloy blade given the alloy's characteristics and the batch size of ~20,000 units. Cold closed-die forging is preferable over hot closed-die forging for Aluminum 7475-T7351 in manufacturing outdoor ceiling fan blades due to its superior dimensional accuracy, surface finish, and cost efficiency Cold forging operates at or near room temperature, resulting in parts with excellent dimensional control and smooth surface finishes, often eliminating the need for additional machining. This process also enhances material strength through strain

hardening, which is beneficial for applications requiring high fatigue resistance, such as fan blades. Additionally, cold forging reduces energy consumption and production costs by eliminating the need for heating, making it more economical for large-scale production. In contrast, hot forging involves heating the aluminum to high temperatures, which can lead to oxidation, scaling, and less precise tolerances, potentially increasing post-processing requirements and costs. Therefore, to produce high-performance, cost-effective fan blades, cold closed-die forging of Aluminum 7475-T7351 is the more suitable manufacturing method[13].

Figure 8 illustrates the cold closed-die forging process, a precision forming technique for shaping Aluminum 7475-T7351 at room temperature. The process begins with a pre-cut workpiece positioned between two dies each machined with a negative imprint of the final blade shape. Initially, the dies remain open (left side of the image); once aligned, the rams drive them inward simultaneously (right side, arrows), applying high pressure to force the material into the die cavities. Excess metal escapes through flash gutters, ensuring dimensional control. A critical advantage of this method is grain-flow alignment: the metal's microstructure follows the blade's aerodynamic profile, enhancing strength and fatigue resistance—unlike machining, which severs the grain and creates weak points. The blades are forged in an annealed state, then heat-treated to T7351 temper for optimal mechanical properties (ductility, impact toughness, and fatigue strength). While alternatives like casting or stamping were considered, they lacked the precision, strength, and consistency required. Closed-die forging achieves near-net shapes with minimal post-processing (e.g., drilling holes), making it ideal for mass-producing high-performance blades. [5].

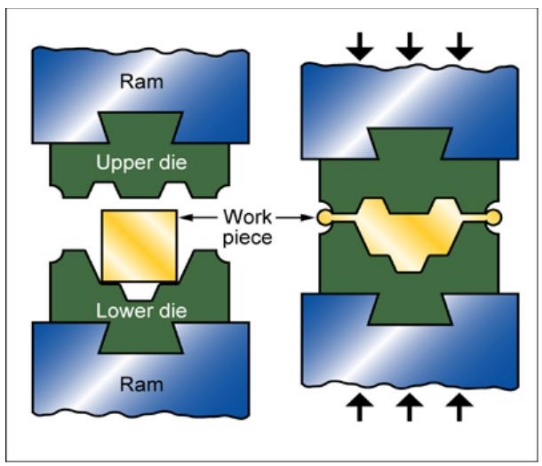


Figure 8. Process Schematics of Cold Closed Die Forging Process [5]

For Aluminum 7475-T7351, mechanical fastening is the most suitable joining method, particularly for attaching the blade to the motor hub or mounting bracket. This is because riveting, bolting, or using threaded inserts allows for non-destructive assembly, preserves material properties, and simplifies maintenance or replacement. Also, mechanical fastening is widely used in aerospace and structural applications involving 7000-series alloys, including 7475, for the same reasons welding is avoided due to loss of strength and risk of cracking. Hence, For Aluminum 7475-T7351 fan blades, mechanical fastening (e.g., bolts, screws, or rivets) is the safest, most effective joining method as it maintains the alloy's full mechanical performance, supports high-volume assembly, and aligns with aerospace-standard practices for joining high-strength, non-weldable aluminum alloys.

While Aluminum 7475-T7351 does not require finishing to maintain its base corrosion resistance, light anodizing, surface cleaning, and optional protective coatings are recommended to enhance aesthetics, weather resistance, and longevity, especially for an outdoor ceiling fan[9].

Conclusion

In conclusion, the final design of the outdoor ceiling fan blade is decided to Aluminum 7475 - T7351 wrought, formed via closed-die forging, because of a thorough evaluation of material and process options. This choice was driven by the alloy's high strength-to-weight performance, stiffness, and fatigue resistance, which far exceed those of the initial ASA plastic design. By transitioning from ASA a UV-resistant plastic[15] to a forged aluminum blade, the design gains significant performance improvements: the metal blade can be made thinner and still handle higher rotational stresses without bending, translating to more efficient airflow delivery. Metal blades are known to be the most efficient at moving air.

Currently, ceiling fan blades are commonly manufactured using materials such as ABS plastic and various metals, including aluminum [10]. ABS plastic blades are lightweight and resistant to moisture but may lack the structural strength required for high-performance applications. Since the cost of the fan blade is expensive ($850), The selection of Aluminum 7475-T7351, is reasonable and elevates the design beyond the typical industry standard by providing superior fatigue resistance and structural integrity and without affecting cost a lot. This ensures that the fan blades maintain their performance and appearance over time, even in challenging outdoor conditions.As a metal, aluminum is inherently resistant to ultraviolet (UV) radiation. Unlike polymers, it does not degrade or lose mechanical properties when exposed to UV light. Therefore, no additional surface treatment is required for UV protection[12].

Appendix

Load Calculation

The blade was considered as a 3D prismatic panel, it's length of the blade is 30 inches [2], and assuming width as 9 inches and taking thickness as 0.5 inches more than what average metal blades would have, due to polymers having low stiffness. The mass of each blade is assumed to be 0.5kg from [9] which has blade made up of ABS, since ASA and ABS have similar range of density, so the mass of blade should be same for same geometry. From the dimensions, we have and RPM as 180 from [2], maximum centrifugal force can be calculated that each blade will have during its operation using formula:

$$F = mr\omega^2$$

where:

- $\omega = 2\pi N = 2 * \pi * \frac{180}{60} = 18.85\ s^{-1}$
- r = 30 inches = 0.762 m
- m = 0.5 kg

$$Fc = 0.5 * 0.762 * 18.85^2$$

$$Fc = 0.5 * 0.762 * 355.31 = 135.36\ N$$

From this calculated centrifugal force of approximately 135.36 N per blade, a factor of safety (FOS) of 2 is considered in ordered to ensure no cracking in. Additionally, a standard deflection value of 2.03 mm from [16], analytical calculation of PVC (a material with similar mechanical properties to ASA) determined this deflection value.

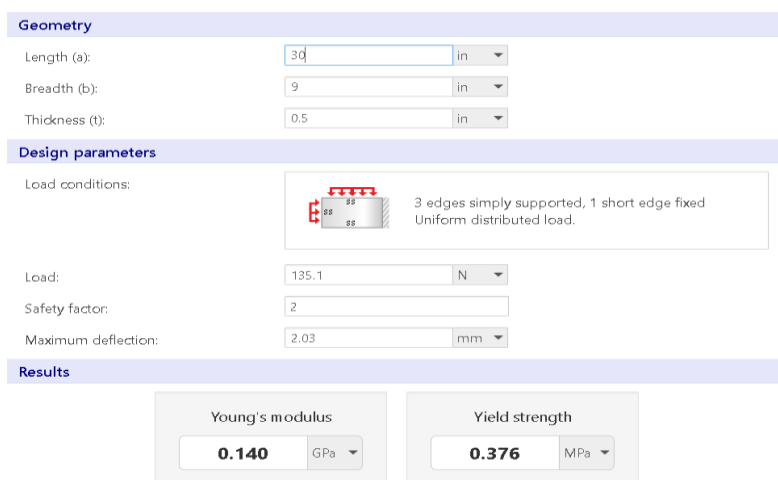


Figure 9. Solver Input for values young's modulus and yield strength

Yield Strength & Modulus

By modeling the fan blade as a panel in bending with one short edge fixed and three edges simply supported, the minimum required Yield Strength and Young's Modulus for the material under normal service conditions were determined using Ansys Granta Solver. Inputs were given for the load of 135.1 N, a safety factor of 2, and a maximum allowable deflection of 2.03 mm, the solver calculated (Figure 9) a minimum yield strength of 0.376 MPa and a Young's modulus of 0.140 GPa. These values establish the material constraints for redesigning the fan blade to prevent structural failure and excessive deflection while maintaining operational integrity.

Fracture Toughness

The failed material used in the recalled fan blade was reported to be ASA (Acrylonitrile styrene acrylate) [3]. As the mode of failure potential crack in fan blade i.e., fracture of the fan blade material the new selection should have a fracture toughness exceeding that of the original component. Relevant properties of ASA are presented in the table below. Minimum fracture toughness for the new material will be set at the mean value for ASA of 1.52 MPa-m1/2 [Table 7].

Table 7. Fracture Toughness and Price for ASA [2]

Properties of Acrylonitrile Styrene Acrylate	
Fracture Toughness	1.52 MPa-$m^{1/2}$
Material Price	2.16 – 2.32 $/kg

Material Price

From (Table 7), the cost per kg of ASA is $2.16 - $2.32 [5]. While a significant increase in material cost would be undesirable for manufacturing, the fan blade is a critical component of the product's overall performance and safety. Given that the recall was due to blade failure, selecting a more robust material is a priority. A moderate price increase is acceptable if it enhances durability, and environmental stability. Additionally, ensuring compatibility with existing manufacturing processes, such as injection molding (for plastic blades) or casting/forging (for metal) blades is essential. Therefore, a cap on material price is set at $5/kg.

Service Temperature

Outdoor ceiling fans are subjected to a wide range of environmental conditions, such as direct sunlight, high humidity, and varying temperatures. These fans are often installed in semi-exposed areas like patios or covered outdoor spaces, where climate control is limited. Since the product was distributed across the U.S. and Canada, the service temperature range was determined based on the extreme temperatures recorded in these regions during 2024. The highest temperature, approximately 40°C, was reported in Las Vegas, while the lowest, around -40°C, occurred in North Dakota[15],.

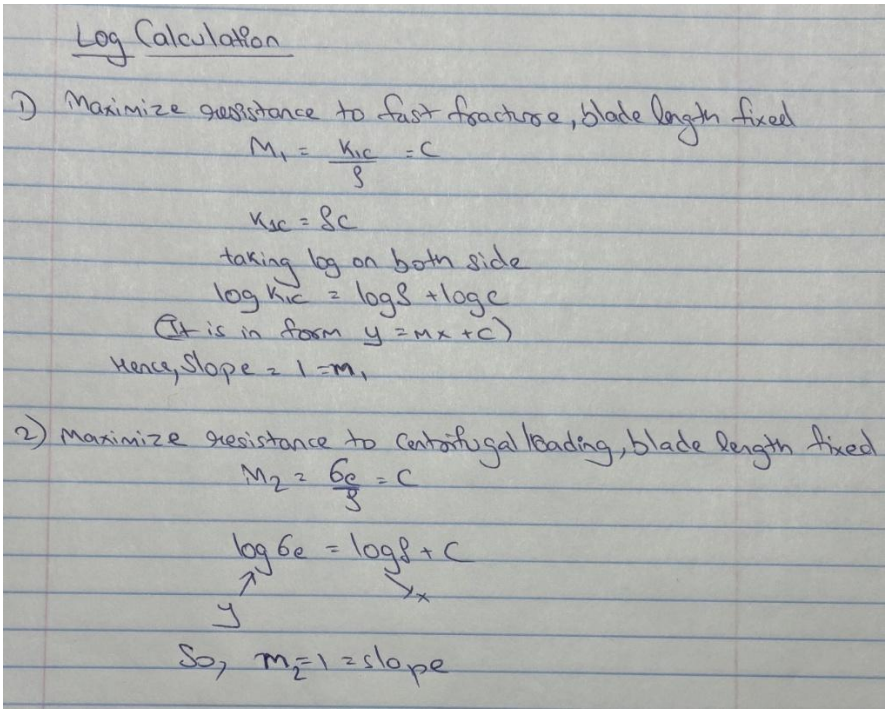
Durability:

Outdoor ceiling fan blades must withstand prolonged exposure to moisture, As the primary material for fan blade (ASA) offers both UV stability and weather resistance without compromising its strength, and potential impacts from any debris [14]. So, material with durability against water and humidity is also crucial to prevent material degradation, while resistance to UV exposure ensures that the blades do not become brittle over time. While durability is not a primary factor in initial material selection, the final choice must meet these requirements through inherent material properties or protective coatings, ensuring long-term performance and safety.

Manufacturing

Manufacturing processes are selected primarily based on material compatibility, shape compatibility, and economic factors. As fan blade in rather simple geometry, and in the recall notice [1], it was given that the total number of recall units were 6400 units in the US and 200 in Canada. Since each unit has 3 blades, that makes total of 19,200 units of total blades required to produce. Hence, the batch size was considered 20,000 units to have some allowance for defects, replacements, and potential market demand. Given this batch size, the manufacturing process should be cost-effective while ensuring structural integrity and durability.

Log Calculations



Log Calculation

1) Maximize resistance to fast fracture, blade length fixed

$$M_1 = \frac{K_{IC}}{\rho} = C$$

$$K_{IC} = \rho C$$

taking log on both side

$$\log K_{IC} = \log \rho + \log C$$

(It is in form $y = mx + c$)

Hence, Slope $= 1 = m_1$

2) Maximize resistance to centrifugal loading, blade length fixed

$$M_2 = \frac{\sigma_e}{\rho} = C$$

$$\underbrace{\log \sigma_e}_{y} = \underbrace{\log \rho}_{x} + C$$

So, $m_2 = 1 =$ slope

Figure 9. Hand Calculations for selection lines using material index (M1 & M2)

Reference

1. U.S. Consumer Product Safety Commission, 2024, "Visual Comfort & Co. Recalls Maverick Coastal Outdoor Ceiling Fans Due to Impact Injury Hazard," [Online]. Available: https://www.cpsc.gov/Recalls/ 2024/Visual-Comfort-Co.-Recalls-Maverick-Coastal-Outdoor-Ceiling-Fans-Due-to-Impact-Injury-Hazard

2. Danielle, C., 2023, "Ceiling Fan Blades: Material, Types and Design," Warmiplanet. [Online]. Available: https://warmiplanet.com/blogs/news/a-guide-to-ceiling-fan-blades

3.Visual Comfort & Co., 2024, "Maverick Coastal 60 Ceiling Fan in Midnight Black with Midnight Black Blades," [Online]. Available: https: //www.visualcomfort. com/media/docs/spec_sheet/ 3/m/ss_3mgmr60mbkmbk.pdf

4. Ansys Inc, 2021, "Ansys Granta EduPack." www.ansys.com/materials

5. Melito, S., 2023, "5 UV Resistant Plastics for Part Designers | Fictiv," Fictiv. [Online]. Available: https://www.fictiv.com/articles/the-top-five-uv-resistant-plastics-for-part-designers.

6. Talreja, R., 2023, "Failure of Unidirectional Fiber Reinforced Composites: A Case Study in Strength of Materials," *Mechanics of Composite Materials*, 59(2), pp. 173–190. DOI: 10.1007/s11029-023-10091-0.T. Hotes,

7. "Selecting the Right Fan Blade Material for the Job - Part 1 - Technically Speaking - Fantech." [Online]. Available: https://www.systemaire.com.au/news.selecting-the-right-fan-blade-material-for-the-job-part-1

8. Gw, 2025, "Choosing the Right Ceiling Fan Blade Material: Metal, Plastic, or Wood?" Dadao. [Online]. Available: https://dadaoenergy.com/blog/the-best-material-for-fan-blades/

9. Bieloborodko, A., 2024, "Choosing the Right Blade Material for Your Ceiling Fan: Factors to Consider," Todays Fans, [Online]. Available: https://www.todaysfans.com/blogs/blog/choosing-the-right-blade-material-for-your-ceiling-fan-factors-to-consider

10. 2020, "7475-T7351 Aluminum :: MakeItFrom.Com." [Online]. Available: https://www.makeitfrom.com/material-properties/7475-T7351-Aluminum.

11. "7475 T7351 Aluminum Plate Sheet | Haomei Aluminum." [Online]. Available: https://www.aerospace-aluminum.com/blog/7475-t7351-aluminum/

12. Jack, 2023, "Ceiling Fan Manufacturing Process: From Design to Assembly," Warmiplanet Ceiling Fan. [Online]. Available: https://ceilingfanmfg.com/blog/ceiling-fan-manufacturing-process/

13. Search, I. Q., "Cold Forging: What Is It? Benefits, Process, Hot vs Cold, Types, Considerations." [Online]. Available: https://www.iqsdirectory.com/articles/forging/cold-forging.html.

14. "ASA's Role in Outdoor Applications Through Weather Resistance," Polymer Resins, [Online]. Available: https://prlresins.com/asas-role-in-outdoor-applications-through-weather-resistance/

15. 2024, "U.S. Climate Summary for January 2024," NOAA Climate.gov. [Online]. Available: https://www.climate.gov/news-features/understanding-climate/us-climate-summary-january-2024.

16. Dhurvey, P., Pathak, N. K., Soni, S., and Pandey, A., 2018, "Aerodynamic and Structural Analysis of Ceiling Fan for Different Blade Materials," Materials Today Proceedings, 5(9), pp. 19604–19613. https://doi.org/10.1016/j.matpr.2018.06.323